What is AI?

measure success in term of fidelity to human performance

thinking humanly

- to make computer think with minds in the full and literal sense.
- associate with human thinking such as decision-making, learning, problem solving, ...

acting humanly

- creating machines that perform functions that require intelligence when performed by human
- study of how to make computers do things which at the moment, human are better

measure against an ideal performance

thinking rationally

- study of mental faculties through the use of computational models
- study of the computations that make it possible to percieve, reason, and act

acting rationally

- computional intelligence is the study of the design of intelligent agents
- AI is concerned with intelligent behavior in artifacts

Acting humanly: The turing test approach

Design to provide a satisfactory operational definition of intelligence.

A computer passes the test if human interrogator, after posing some written questions. Interrogator couldn't tell whether the responses come from who. The computer would need to posses the following capabilities:

- natural language processing to communicate successfully in English.
- knowledge representation to store what it knows or hears.
- automated reasoning to use knowledge presentation to answer questions and draw new conclusions.
- machine learning

Total turing test includes a video signal so that interrogator can test the subject's perceptual abilities, to pass the computer will need:

- computer vision to perceive objects
- robotics to manipulate objects and move about

Thinking humanly: the cognitive modeling approach.

to get inside the actual working of human minds, there are 3 ways:

- **through introspection** - trying to catch our own thoughts as they go by
- **through psychological experiments** - observing a person in action
- **through brain image** - observing the brain in action

PEAS

Perfomance measure - define the success of an agent

Environment - surrounding of an agent at every instant

Actuator - part of the agent that delivers the output of an action to the environment

Sensor - receptive parts of an agent which takes in the input for the agent.

agent	performance measure	environment	actuator	sensor
automated car drive	comfortable, safety, maximum distance	roads, traffic, vehicles	steering wheel, brake, mirror, accelerator	camera, gps, odometer

ENVIRONMENT & CHARACTERISTICS

observable

- fully observable - has access to all required information to complete task.
- partially observable - partial information to solve AI problems.

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- Deterministic - determined base on specific state, it ignore uncertainty
- Stochastic

Epicsodic Vs. Sequential

- Epicsodic - one-shot actions
- Sequential - requires memory of past actions to determine next best actions.

Static Vs. Dynamic

- static - rely on data-knowledge that doesn't change frequently
- dynamic - rely on data-knowledge that change frequently

Discrete Vs. Continuous

- Discrete - finate number of percepts and actions
- Continuous - rely on unknown and rapidly changing data sources.

Agents

- Single agent - 1 agent is involve and operating by itself
- Multi agent - multi agent involve and operating